

1, Explain uses of ping command in network ?

- \* Used to verify the network connection establishment.
- \* Used to verify the host device is reachable or not.
- \* Used to verify know the data loss in the network.
- \* Used to ~~know~~ test the DNS.
- \* Helps to troubleshoot network problems.

2, Explain netbios protocol ?

- \* It is a naming service.
  - \* Helps to communicate using names instead of IP address.
- Eg: ping 192.168.1.2 → traditional  
ping OFFICE-PC → NetBIOS

3, Explain SMB protocol ?

- \* SMB - Server Message Block.
- \* It allows to create "Share Folders", "Access Remote Storage".

Eg: File Server

- \* Commonly used in Local Office N/w.

4, Explain ping process from 50.0.0.10 to 50.0.0.20 ?

- \* Ping generates a dummy data and this data is transferred through the 7 layers of OSI model.
- \* In Application layer, dummy data is known as "Echo", it has size of 32 bytes with an ICMP protocol.
- \* In Presentation Layer format of the data is defined.
- \* Session layer defines the time frame and Transport layer uses several protocols.
- \* In Network Layer, Source and Destination IP is defined with the help of Internet Protocol and this is known as "Packets".
- \* In Data Link Layer, Source MAC and Destination MAC will be defined and Ethernet Protocol is responsible but Destination MAC is unknown.
- \* So Ethernet protocol call Address Resolution Protocol (ARP). ARP creates a request (SIP, SMAC, DIP, DMAC:00) with SMAC and DMAC as FF:FF.
- \* ARP reply consists of the (SIP, SMAC, DIP, DMAC) and it stores the value in ARP table and then shares to the Ethernet protocol.
- \* In Physical layer, the data are converted into Bits and transferred as Electrical Signals through the cables and reaches the target IP device.